

Exact formulas for partial sums of the Möbius function expressed by partial sums weighted by the Liouville lambda function

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High-level overview and takeaways of the talk

- ▶ Study new expressions for partial sums of a signed classical function
- ▶ Identify new unsigned sequences through which we can express these partial sums, or summatory functions
- ▶ Try to keep things in perspective at a high level

Definitions and notation

- ▶ The function $\omega(n)$ (and $\Omega(n)$) counts the number of distinct prime factors of any n without (and with, respectively) multiplicity.
- ▶ Recall that the **Möbius function** is defined as

$$\mu(n) = \begin{cases} 1, & \text{if } n = 1; \\ (-1)^{\omega(n)}, & \text{if } \omega(n) = \Omega(n) \text{ (i.e., if this } n \geq 2 \text{ is squarefree);} \\ 0, & \text{otherwise.} \end{cases}$$

- ▶ The summatory function given by the **Mertens function** is defined as follows:

$$M(x) := \sum_{n \leq x} \mu(n), \text{ for } x \geq 1.$$

- ▶ Related functions include the **Liouville lambda function**, $\lambda(n) := (-1)^{\Omega(n)}$ for $n \geq 1$, and its partial sums $L(x) := \sum_{n \leq x} \lambda(n)$ for $x \geq 1$.

Definitions of auxiliary unsigned functions

- ▶ We fix the notation for the Dirichlet inverse function (inverse taken with respect to the operation of Dirichlet convolution, e.g., $(f * h)(n) = \sum_{d|n} f(d)h\left(\frac{n}{d}\right)$) as follows:

$$g(n) := (\omega + \mathbb{1})^{-1}(n), \text{ for } n \geq 1.$$

- ▶ We define the partial sums for $x \geq 1$ as

$$G(x) := \sum_{n \leq x} g(n) = \sum_{n \leq x} \lambda(n) |g(n)|.$$

- ▶ Where did the definition of $g(n)$ come from? Its partial sums are related to the classical prime counting function by

$$\chi_{\mathbb{P}}(n) + \delta_{n,1} = (\omega + \mathbb{1}) * \mu(n), n \geq 1,$$

by Möbius inversion since

$$\omega(n) = \sum_{p|n} 1 = \sum_{d|n} \chi_{\mathbb{P}}(d).$$

New explicit formulas for $M(x)$

Theorem

For all $x \geq 1$

$$(1a) \quad M(x) = G(x) + \sum_{1 \leq k \leq x} |g(k)| \pi \left(\left\lfloor \frac{x}{k} \right\rfloor \right) \lambda(k),$$

$$(1b) \quad M(x) = G(x) + \sum_{1 \leq k \leq \frac{x}{2}} \left(\pi \left(\left\lfloor \frac{x}{k} \right\rfloor \right) - \pi \left(\left\lfloor \frac{x}{k+1} \right\rfloor \right) \right) G(k),$$

$$(1c) \quad M(x) = G(x) + \sum_{p \leq x} G \left(\left\lfloor \frac{x}{p} \right\rfloor \right).$$

Remarks on the significance of the new formulas for $M(x)$ in terms of $G(x)$

- ▶ The summands are sign-weighted by $\lambda(n)$ with unsigned magnitudes that have “nicer” properties.
- ▶ For comparison, we have the less predictably signed expansion:

$$(2) \quad M(x) = \sum_{d \leq \sqrt{x}} \mu(d) L\left(\left\lfloor \frac{x}{d^2} \right\rfloor\right), \text{ for } x \geq 1.$$

- ▶ We argue that the unsigned summands in the previous theorem are “nicer” than classical expansions of $M(x)$ such as the one in (2)?

Properties of the unsigned sequences

- For all $n \geq 1$, $\text{sgn}(g(n)) = \lambda(n)$
- An exact expression is given by:

$$\lambda(n)g(n) = \sum_{d|n} \mu^2\left(\frac{n}{d}\right) C_{\Omega}(d), n \geq 1,$$

where

$$C_{\Omega}(n) = \begin{cases} 1, & \text{if } n = 1; \\ (\Omega(n))! \times \prod_{p^{\alpha} || n} \frac{1}{\alpha!}, & \text{if } n \geq 2. \end{cases}$$

- For all squarefree integers $n \geq 1$, we have that

$$|g(n)| = \sum_{m=0}^{\omega(n)} \binom{\omega(n)}{m} \times m!$$

Average order asymptotics

- There is an absolute constant $B_0 > 0$ so that

$$\frac{1}{n} \times \sum_{k \leq n} \log C_\Omega(k) = B_0 \cdot (\log \log n)(\log \log \log n)(1 + o(1)).$$

- The average order of $\log |g(n)|$ is given by

$$\frac{1}{n} \times \sum_{k \leq n} \log |g(k)| = \left(\frac{B_0}{2} \cdot (\log \log n)(\log \log \log n) - \frac{1}{2} \log \left(\frac{\pi^2}{6} \right) \right) (1 + o(1)).$$

- The variance of $\log C_\Omega(n)$ (and $\log |g(n)|$ up to a constant factor) is

$$\sigma_\Omega(x) = D_0 \sqrt{x} (\log \log x)(\log \log \log x)(1 + o(1)),$$

for $D_0 > 0$ an absolute constant.

Properties of the unsigned sequences (cont'd)

- Recall the Erdős-Kac theorem:

$$\frac{1}{x} \times \# \left\{ 2 \leq n \leq x : \frac{\omega(n) - \log \log x}{\sqrt{\log \log x}} \leq z \right\} = \Phi(z) + o(1).$$

- In analog, we conjecture that there are absolute constants $B_0, D_0 > 0$ such that for $z \in \mathbb{R}$ as $x \rightarrow \infty$

$$\frac{1}{x} \times \# \left\{ 2 \leq n \leq x : \frac{\log C_\Omega(n) - B_0 \cdot (\log \log x)(\log \log \log x)}{D_0 \cdot (\log \log x)(\log \log \log x)} \leq z \right\} = \Phi(z) + o(1).$$

- If the previous conjecture holds, then for any $y > 0$

$$\begin{aligned} \frac{1}{x} \times \# \left\{ 2 \leq n \leq x : -y \leq |g(n)| - \frac{1}{n} \times \sum_{k \leq n} |g(k)| \leq y \right\} \\ = \Phi \left(\frac{\log \left(\frac{\pi^2 y}{6} \right) - B_0 \cdot (\log \log x)(\log \log \log x)}{D_0 \cdot (\log \log x)(\log \log \log x)} \right) + o(1). \end{aligned}$$

Conclusions – Taking a step back – What we've done

- ▶ We defined $g(n) := (\omega + \mathbb{1})^{-1}(n)$ as the shifted Dirichlet inverse of the strongly additive function, $\omega(n)$.
- ▶ We precisely connected $C_\Omega(n)$ to $g(n)$ and used it to prove formulas for the average orders of the unsigned sequences.
- ▶ We have conjectured a limiting CLT for the distribution of $\log C_\Omega(n)$ (and so $|g(n)|$) for $n \leq x$ as $x \rightarrow \infty$.
- ▶ We connected the Mertens function $M(x)$ with the partial sums $G(x) := \sum_{n \leq x} \lambda(n) |g(n)|$ via exact formulas for all $x \geq 1$.

Conclusions

The End

Questions?

Comments?

Feedback?

Thank you for attending!

n	n	$g(n)$	$\frac{\sum_{d n} C_{\Omega}(d)}{ g(n) }$	$G(x)$
2	2^1	-2	1.000	-1
3	3^1	-2	1.000	-3
4	2^2	2	1.500	-1
5	5^1	-2	1.000	-3
6	$2^1 3^1$	5	1.000	2
7	7^1	-2	1.000	0
8	2^3	-2	2.000	-2
9	3^2	2	1.500	0
10	$2^1 5^1$	5	1.000	5
11	11^1	-2	1.000	3
12	$2^2 3^1$	-7	1.286	-4
13	13^1	-2	1.000	-6
14	$2^1 7^1$	5	1.000	-1
15	$3^1 5^1$	5	1.000	4
16	2^4	2	2.500	6
17	17^1	-2	1.000	4
18	$2^1 3^2$	-7	1.286	-3
19	19^1	-2	1.000	-5
20	$2^2 5^1$	-7	1.286	-12
21	$3^1 7^1$	5	1.000	-7
22	$2^1 11^1$	5	1.000	-2
23	23^1	-2	1.000	-4
24	$2^3 3^1$	9	1.556	5
25	5^2	2	1.500	7
26	$2^1 13^1$	5	1.000	12
27	3^3	-2	2.000	10

n	n	$g(n)$	$\frac{\sum_{d n} C_{\Omega}(d)}{ g(n) }$	$G(x)$
28	$2^2 7^1$	-7	1.286	3
29	29^1	-2	1.000	1
30	$2^1 3^1 5^1$	-16	1.000	-15
31	31^1	-2	1.000	-17
32	2^5	-2	3.000	-19
33	$3^1 11^1$	5	1.000	-14
34	$2^1 17^1$	5	1.000	-9
35	$5^1 7^1$	5	1.000	-4
36	$2^2 3^2$	14	1.357	10
37	37^1	-2	1.000	8
38	$2^1 19^1$	5	1.000	13
39	$3^1 13^1$	5	1.000	18
40	$2^3 5^1$	9	1.556	27
41	41^1	-2	1.000	25
42	$2^1 3^1 7^1$	-16	1.000	9
43	43^1	-2	1.000	7
44	$2^2 11^1$	-7	1.286	0
45	$3^2 5^1$	-7	1.286	-7
46	$2^1 23^1$	5	1.000	-2
47	47^1	-2	1.000	-4
48	$2^4 3^1$	-11	1.818	-15
49	7^2	2	1.500	-13
50	$2^1 5^2$	-7	1.286	-20
51	$3^1 17^1$	5	1.000	-15
52	$2^2 13^1$	-7	1.286	-22
53	53^1	-2	1.000	-24